

IIT BOMBAY
CHEMICAL ENGINEERING

CL420: Tutorial Problems

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Mass Balances: Stoichiometry

The major elemental demands of a growing cell are carbon, oxygen, hydrogen, and nitrogen. Assuming the carbon source to be characterized by (CH_2O) and nitrogen by NH_4^+ , growth follows the general balance below.



(new cellular material shown as $\text{C}_a\text{H}_b\text{O}_c\text{N}_d$)

(a) Write down the restriction among the coefficients α , β , ... from consideration of elemental conservation. Given the “composition” of a cell, what additional information is needed to solve for the coefficient values?

(b) Repeat part (a) for photosynthetic systems:

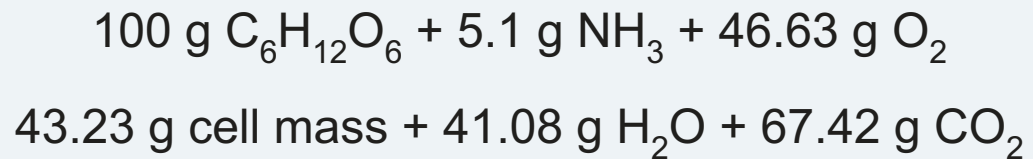


(c) Describe an explicit series of experiments by which you would evaluate the stoichiometric formula of the cell (a , b , c , d).

Stoichiometry: Aerobic vs. Anaerobic Species

To conveniently calculate nutrient consumption, aeration, and heat-transfer rates, a general stoichiometric equation can be formulated by assigning the cell an apparent molecular formula.

(a) Calculate the cell formula for growth of yeast on sugar, given the following balance (1 mol of cell produced):



(b) If the above reaction produces 0.30017 “mol” of cell, what cell formula results?

Note: elemental (C, H, O, N) and mass balances on the part-(a) reaction give the mole quantities of each species; dividing by 0.30017 mol cell normalizes the empirical formula $C_aH^bO^cN^d$ to one mole of cell.

(c) Anaerobic fermentations also yield partially oxygenated by-products. Keeping the cell formula from (b), find the unknown coefficients (mole quantities) below:

0.55556 glucose + α ammonia

→ β cell mass + 0.05697 glycerol

+ γ butanol (alcoholic product)

+ ε succinic acid (acid product)

+ 0.01164 water + 1.03076 carbon dioxide

+ 1.00380 ethanol

Coefficients α , β , γ , and ε are the unknowns to be solved by elemental balance

Network Mass Balance

You have a friend who asked to run a fermentation in a laboratory course but neglected to weigh the glucose he added and forgot to analyze for ethanol.

He has found that the fermentation produced the compounds shown in Table. He knows that all glucose ($C_6H_{12}O_6$) is oxidized to pyruvic acid ($CH_3COCOOH$) by the EMP pathway, and that no other products except ethanol are formed.

He comes to you for help in salvaging something out of this mess. **How many moles of ethanol should you tell him have been formed?** *(State your assumptions.)*

TABLE : Compounds and

Compound	Moles
Lactic acid ($CH_3CHOHCOOH$)	10
Acetic acid (CH_3COOH)	5
Carbon dioxide (CO_2)	15
Hydrogen (H_2)	10

All quantities in the table are given in moles. Ethanol was not measured, and glucose consumed was not weighed — both must be recovered from the EMP stoichiometry and an elemental (C, H, O) balance over the network.